Section 1 — Incident Header

Incident ID: INC-20260421-006
Date: 21-Apr-2026
Time Opened (IST): 22:14 IST
Severity: Critical
Client: ManuTech Industries
Industry: Manufacturing
L1 Analyst: Priya Nair
L2 Analyst: Rohan Desai
L3 Analyst: Nisha Kapoor
Current Status: Active – Escalated to L2 Investigation
Sentinel Incident Number: 457892

Section 2 — Client & Environment Context

ManuTech Industries is a mid-to-large scale manufacturing organization specializing in industrial automation components used across automotive and heavy engineering sectors. Headquartered in Pune, the company operates three production plants across Maharashtra and Gujarat, with a workforce of approximately 2,800 employees. Their operations are highly dependent on uninterrupted production lines, automated machinery control systems, and tightly integrated IT-OT environments.

The organization maintains a hybrid identity infrastructure, with an on-premises Active Directory environment synchronized to Azure AD using Azure AD Connect. All employees authenticate through Azure AD, with Conditional Access enforcing MFA for remote access and privileged roles. The endpoint estate consists of roughly 1,900 Windows 10/11 workstations and 140 Windows Server 2019/2022 systems, including domain controllers, file servers, and manufacturing execution system (MES) servers.

ManuTech runs Microsoft 365 E5 licensing, giving them full deployment of Microsoft Defender for Endpoint, Defender for Identity, Defender for Office 365, and Microsoft Purview. Their Azure footprint includes production-support workloads such as telemetry ingestion, IoT device data processing, and backup storage accounts.

The network architecture is segmented but interconnected — plant networks connect back to HQ via site-to-site VPN tunnels, and remote engineers use split-tunnel VPN through Azure VPN Gateway. Critical assets include MES servers, domain controllers, and production scheduling databases. Any disruption to these systems directly impacts manufacturing output.

From a compliance standpoint, ManuTech adheres to ISO 27001 and sector-specific industrial cybersecurity guidelines. Notably, they recently completed a quarterly backup validation exercise — ensuring shadow copies and backup catalogs are intact and recoverable.

This context is critical: any activity targeting system recovery mechanisms, such as shadow copy deletion, is not just suspicious — it is a direct precursor to ransomware impact in an environment where downtime translates immediately to financial loss and production halt.

Section 3 — Attack Narrative

The attacker had already established a quiet foothold inside the network hours earlier, long before any visible disruption began. Their entry point had not triggered immediate alarms — likely a compromised endpoint obtained through earlier credential theft or a malicious payload execution that blended into normal user activity. They moved carefully at first, observing, mapping, and waiting.

At 21:52 IST, the attacker shifted intent.

They opened a command shell on the compromised workstation — a mid-level engineering machine that had access to shared drives and internal systems. Their objective was no longer reconnaissance. It was preparation.

The first command they executed was simple:
net.exe view /domain
A quick enumeration of available systems. The response confirmed what they needed — domain visibility, reachable servers, and the scale of the environment.

Minutes later, they escalated their actions.

They invoked **vssadmin.exe**, targeting shadow copies — the silent safety net of Windows systems.
vssadmin delete shadows /all /quiet

No prompts. No confirmations. The command executed silently, wiping out system restore points that could later be used to recover encrypted files.

But they didn’t stop there.

They followed immediately with:
wbadmin delete catalog -quiet

This ensured that even backup catalogs — the index of recoverable backups — were rendered useless. Recovery was being systematically dismantled, layer by layer.

The attacker understood that partial disruption wasn't enough. Full impact required certainty.

At 22:08 IST, they executed another command:
bcdedit /set {default} recoveryenabled No

This disabled Windows recovery options at boot. Even if the system attempted to recover after failure, it would have no path to do so.

The sequence was deliberate. Not exploratory. Not experimental. Each command was part of a known ransomware playbook — remove recovery, remove backups, remove fallback.

What remained was a system with no safety net.

The attacker paused.

They did not trigger encryption yet.

Instead, they waited — ensuring their changes had taken effect across the system. Watching. Confirming. Preparing for the final stage, where files would be locked and systems rendered unusable.

By the time the SOC would notice, the groundwork for impact had already been laid.

Section 4 — Detection & Alert

The first signal did not arrive as a single, obvious alarm. It came in fragments.

At 22:12 IST, Microsoft Sentinel generated a Medium severity alert titled:

"Suspicious System Recovery Modification — vssadmin Execution"

Two minutes later, another alert appeared:

"Backup Catalog Deletion via wbadmin"

Then a third:

"System Recovery Disabled via bcdedit"

All three alerts referenced the same device: **MT-ENG-WS-047**.

The analytics rule responsible for the initial alert was a custom Sentinel rule monitoring destructive system recovery commands:

```
DeviceProcessEvents
| where TimeGenerated > ago(30m)  // Look for recent activity
```

```
| where ProcessCommandLine has_any ("vssadmin delete shadows", "wbadmin delete catalog", "bcdedit /set")  // Known ransomware precursor commands
| where InitiatingProcessAccountName !in ("SYSTEM", "TrustedInstaller")  // Exclude system-level maintenance activity
| summarize CommandCount = count() by DeviceName, InitiatingProcessAccountName, bin(TimeGenerated, 5m)
| where CommandCount >= 1  // Trigger on any occurrence due to high risk nature
```

The specific values that triggered the rule were:

- **DeviceName:** MT-ENG-WS-047
- **InitiatingProcessAccountName:** j.patil
- **ProcessCommandLine:**
    - vssadmin delete shadows /all /quiet
    - wbadmin delete catalog -quiet
    - bcdedit /set {default} recoveryenabled No
- **Time Window:** 21:58 IST – 22:10 IST

Individually, each alert was classified as Medium severity. But Sentinel's incident grouping logic correlated them into a single incident due to shared entity context — same device, same user, close time proximity.

By 22:14 IST, the Incidents queue displayed:

- **Title:** Multiple System Recovery Destruction Commands Detected
- **Severity:** High (auto-correlated)
- **Entities:**
    - Device: MT-ENG-WS-047
    - Account: j.patil
- **Time:** 22:14 IST

Seconds later, a fourth alert appeared:

"Network Enumeration via net.exe"

Now four alerts, one host, within 22 minutes.

The pattern was no longer ambiguous. It was forming into something structured — something deliberate.

That was the moment it surfaced in Priya's queue.

Section 5 — L1 Triage

Priya heard the alert before she saw it.

A soft but distinct tone echoed across the SOC floor — not the sharp escalation tone reserved for Critical alerts, but enough to pull her attention away from the low-severity noise she had been clearing.

Her Sentinel window was already open in full screen. The new incident slid into the top of the queue.

"Multiple System Recovery Destruction Commands Detected — High"

She didn't click immediately.

Her eyes scanned the row first — habit.

Same device repeated across four linked alerts. Same user. Timestamps clustered tightly. That alone made her pause.

"Same host… four different commands…" she murmured under her breath.

She clicked.

The incident panel expanded on the right. She read the alert list one by one — not skimming, but parsing.

- net.exe enumeration
- vssadmin delete shadows
- wbadmin delete catalog
- bcdedit recovery disabled

She stopped at the third one.

Her mind jumped ahead before she finished reading.

Shadow copies gone… backup catalog gone… recovery disabled…

She leaned slightly forward in her chair.

This wasn’t normal admin activity. Even legitimate maintenance didn’t chain commands like this in rapid succession — and definitely not from a user workstation.

She opened the **Entity Behavior** tab for **MT-ENG-WS-047**.

The timeline populated.

She scrolled back 24 hours.

Nothing unusual until the last hour. Then suddenly — a cluster of process executions. All command-line driven. No prior history of administrative tools on this endpoint.

She switched to **Defender for Endpoint** (Tab 3).

Opened the device timeline.

There it was — process tree view.

cmd.exe spawning vssadmin.exe.

Another instance of cmd.exe spawning wbadmin.exe.

She clicked into the command line details.

Exact flags. Quiet mode. No prompts.

She felt that slight moment of hesitation.

Could this be IT doing emergency cleanup? Maybe backup corruption?

She checked the account: **j.patil**.

Not IT.

She flipped back to Sentinel and opened the alert details again. Checked the timestamp spread — 22 minutes.

Then she remembered something from training.

Ransomware doesn’t start with encryption.

It starts by removing recovery.

She sat back for a second, processing.

Four Medium alerts… but together…

This wasn’t Medium.

She didn’t wait for a fifth.

She moved her cursor to the severity field and changed it:

High → Critical

Then opened Microsoft Teams on her side panel — SOC-Incidents channel already visible.

Her fingers moved quickly, but her wording stayed precise.

She typed:

Escalation — Critical Incident (ManuTech Industries)

Multiple correlated alerts on single host MT-ENG-WS-047 within 22-minute window.

Key observations:

- net.exe domain enumeration
- vssadmin delete shadows /all /quiet
- wbadmin delete catalog -quiet
- bcdedit recovery disabled

All commands executed under user j.patil (non-privileged engineering account).

No prior history of admin-level activity on this endpoint. Process tree confirms cmd.exe spawning all commands.

Initial assessment: Strong indicator of ransomware staging (pre-encryption phase — recovery mechanisms being disabled).

Request L2 support to confirm:

- Any file encryption activity initiated (DeviceFileEvents)
- Lateral movement or additional hosts impacted
- Source of initial compromise on MT-ENG-WS-047

Escalating as Critical due to high likelihood of imminent impact.

She hit send.

Then pinged Rohan directly.

The incident clock was already ticking.

Section 6 — L2 Investigation

Rohan read the Teams message before opening anything else.

He always did.

He leaned slightly toward his center monitor, eyes scanning Priya's summary line by line. He wasn't looking for confirmation — he was looking for her thinking.

"Recovery mechanisms being disabled…"

He nodded faintly.

She had seen it.

He opened Microsoft Sentinel.

Pulled up **Incident 457892**.

Before touching any logs, he expanded the **Investigation Graph**.

Device at center: **MT-ENG-WS-047**
User: **j.patil**
Processes branching out: cmd.exe, vssadmin.exe, wbadmin.exe, bcdedit.exe

It already looked like a playbook.

Not noise. Not random.

He opened Notepad++ on his second monitor — his KQL scratchpad.

Typed his first question:

// Question 1: Did the attacker initiate encryption or file modification activity after disabling recovery?

Then wrote the query:

```
DeviceFileEvents
| where TimeGenerated > ago(2h)  // Focus on recent activity window
| where DeviceName == "MT-ENG-WS-047"  // Target affected host
| where ActionType in ("FileCreated", "FileModified", "FileRenamed")  // Look for file changes
| summarize FileEvents = count() by bin(TimeGenerated, 5m), ActionType
| order by TimeGenerated asc
```

He ran it.

The results came back quickly.

Normal file activity until 22:10.

Then a pause.

No spike. No mass rename patterns. No extension changes.

He leaned back slightly.

Good. Not yet.

That meant they were still in the staging phase.

He typed his next question.

// Question 2: What process chain led to execution of destructive commands? Identify initial entry point

```
DeviceProcessEvents
| where TimeGenerated > ago(3h)  // Extend window to trace origin
| where DeviceName == "MT-ENG-WS-047"
| project TimeGenerated, FileName, ProcessCommandLine, InitiatingProcessFileName,
InitiatingProcessCommandLine
| order by TimeGenerated asc
```

The output told a story.

21:14 — powershell.exe executed with encoded command
21:15 — outbound connection initiated
21:16 — cmd.exe spawned

He zoomed into that first PowerShell event.

Copied the command line.

Opened CyberChef on his browser.

Decoded Base64.

The payload resolved into a simple download cradle — pulling a script from an external IP.

He didn’t need to guess anymore.

This wasn’t manual.

This was delivered.

He pivoted again — this time mentally.

Where did the PowerShell come from?

Scrolled further up.

21:12 — winword.exe execution.

He paused.

There it was.

Macro entry point.

Everything lined up.

He started reconstructing the timeline.

Attack Timeline

Timestamp (IST)	**Event**	**Data Source**	**MITRE Technique**
21:12	WINWORD.EXE opened document	DeviceProcessEvents	T1204 — User Execution
21:14	PowerShell encoded command executed	DeviceProcessEvents	T1059.001 — PowerShell
21:15	External IP connection initiated	DeviceNetworkEvents	T1105 — Ingress Tool Transfer
21:52	net.exe enumeration	DeviceProcessEvents	T1087 — Account Discovery
21:58	vssadmin delete shadows	DeviceProcessEvents	T1490 — Inhibit System Recovery
22:05	wbadmin delete catalog	DeviceProcessEvents	T1490 — Inhibit System Recovery
22:08	bcdedit recovery disabled	DeviceProcessEvents	T1490 — Inhibit System Recovery

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He moved to IOC extraction.

IOC Table

IOC	Type	Source	Enrichment Result
185.224.128.77	IP	DeviceNetworkEvents	AbuseIPDB: 92% malicious — known malware hosting
powershell encoded payload	Command	DeviceProcessEvents	Obfuscated download cradle
MT-ENG-WS-047	Device	Endpoint	Compromised host
j.patil	User	Identity	Valid user account, no prior admin behavior

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He opened AbuseIPDB in his right monitor.

Confirmed the IP.

Recent reports. Malware distribution. Command-and-control infrastructure.

He exhaled slowly.

Everything aligned.

This wasn't hypothetical anymore.

This was a ransomware deployment in progress — paused just before detonation.

He switched back to Sentinel.

Started writing containment steps — not generic, not delayed.

Specific.

Containment Recommendations

1. **Immediately isolate device MT-ENG-WS-047 using Microsoft Defender for Endpoint (Device Isolation — Full).**

2. **Block external IP 185.224.128.77 in Microsoft Defender for Endpoint indicator list (Network Indicator — Block).**
3. **Force password reset for user j.patil and revoke all active sessions in Azure AD (SigninLogs validation required post-reset).**
4. **Trigger full endpoint scan via Defender for Endpoint on MT-ENG-WS-047 before rejoining network.**
5. **Review DeviceProcessEvents across environment for similar PowerShell encoded command patterns (possible lateral spread).**
6. **Verify backup integrity on affected systems — specifically shadow copies and backup catalog availability across critical servers.**

He paused before closing his notes.

Looked once more at the timeline.

They had caught it between preparation and impact.

Not early.

But not too late.

He began drafting his full investigation notes — knowing that the next phase wouldn't be about detection anymore.

It would be about prevention of what almost happened.

Section 7 — L3 Detection Tuning

Nisha didn't look at the incident queue.

She never did.

Her screen was already split — left side: Rohan's investigation notes, exported into a structured report. Right side: Microsoft Sentinel Analytics Rules editor.

She read the timeline first.

Her eyes paused at **21:58 IST — vssadmin delete shadows**.

Then moved up.

21:52 — net.exe enumeration

Then further.

21:14 — PowerShell encoded execution

She leaned back slightly.

Detection started too late.

She scrolled to the detection section Rohan had documented. The rule that fired was visible — simple, direct, but isolated.

She opened the Analytics Rule in Sentinel.

Read the KQL slowly, line by line.

Before-State KQL Rule

```
DeviceProcessEvents
| where TimeGenerated > ago(30m)
| where ProcessCommandLine has_any ("vssadmin delete shadows", "wbadmin delete catalog", "bcdedit /set")
| where InitiatingProcessAccountName !in ("SYSTEM", "TrustedInstaller")
| summarize CommandCount = count() by DeviceName, InitiatingProcessAccountName, bin(TimeGenerated, 5m)
| where CommandCount >= 1
```

She didn’t type immediately.

Instead, she opened Notepad++ and wrote one line at the top:

“This rule detects single destructive commands, but fails to correlate multi-stage ransomware behavior across a time window.”

That was the problem.

Each command triggered separately. Medium severity. It was Priya — not the system — that connected the dots.

Nisha turned back to her keyboard.

Started writing the improved logic.

After-State Improved KQL Rule

```
DeviceProcessEvents
| where TimeGenerated > ago(30m)  // Investigation window
| where ProcessCommandLine has_any (
   "net.exe view",
   "vssadmin delete shadows",
   "wbadmin delete catalog",
   "bcdedit /set"
)  // Include enumeration + recovery disabling commands
| where InitiatingProcessAccountName !in ("SYSTEM", "TrustedInstaller")  // Exclude system processes
| summarize
   CommandsExecuted = make_set(ProcessCommandLine),
   CommandTypes = dcount(ProcessCommandLine),
   StartTime = min(TimeGenerated),
   EndTime = max(TimeGenerated)
   by DeviceName, InitiatingProcessAccountName
| where CommandTypes >= 3  // Trigger only when multiple distinct ransomware precursor actions observed
| extend TimeWindow = datetime_diff("minute", EndTime, StartTime)
| where TimeWindow <= 30  // Ensure actions occur within a short timeframe
```

She read it back once.

This time, she visualized the incident.

Four commands. Same host. Same user. Within 22 minutes.

This rule would have triggered once — as a single, high-confidence alert.

Not four Mediums.

Not dependent on human pattern recognition.

She added a final comment in her notes:

“This change shifts detection from atomic events to behavioral correlation — enabling earlier and clearer identification of ransomware staging.”

She saved the draft.

Then leaned back again, just briefly.

This was the difference.

Detection that informs.

Versus detection that waits to be interpreted.

Section 8 — Final Incident Report

Part A — Executive Summary

A security incident was identified on an engineering workstation where unauthorized activity attempted to disable system recovery mechanisms — a known precursor to ransomware attacks. The activity was detected before any file encryption or operational disruption occurred.

The affected system was immediately isolated, preventing further spread. The compromised user account was secured, and the malicious external connection was blocked.

There is no evidence of data loss or production impact.

The incident was contained in its early stage, and improvements have been made to detection systems to ensure faster identification of similar threats in the future.

Part B — Full Technical Summary

Incident ID: INC-20260421-006
Client: ManuTech Industries
Attack Type: Ransomware Precursor Activity (Pre-Encryption Stage)

MITRE ATT&CK Techniques Observed

- T1204 — User Execution
- T1059.001 — PowerShell
- T1105 — Ingress Tool Transfer

- T1087 — Account Discovery
- T1490 — Inhibit System Recovery

Timeline Summary

Time (IST)	Event
21:12	Malicious document opened (WINWORD.EXE)
21:14	PowerShell encoded command executed
21:15	Outbound connection to malicious IP
21:52	Domain enumeration via net.exe
21:58	Shadow copies deleted (vssadmin)
22:05	Backup catalog deleted (wbadmin)
22:08	System recovery disabled (bcdedit)
22:14	Detection and SOC escalation

Impact Assessment

- No file encryption observed
- No lateral movement confirmed
- No data exfiltration detected
- Single endpoint impacted (MT-ENG-WS-047)
- Incident contained before business impact

IOC List

IOC	Type	Source	Enrichment Result
185.224.128.77	IP Address	DeviceNetworkEvents	AbuseIPDB: High confidence malicious (malware hosting)

Encoded PowerShell Command	Command	DeviceProcessEvents	Obfuscated download cradle
MT-ENG-WS-047	Device	Endpoint	Confirmed compromised
j.patil	User Account	Identity	Legitimate user, credentials likely abused

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Containment Actions Taken

- Device MT-ENG-WS-047 isolated via Microsoft Defender for Endpoint
- Malicious IP blocked via Defender Network Indicators
- User account j.patil password reset and sessions revoked via Azure AD
- Full endpoint scan initiated and completed
- Backup systems verified for integrity across environment

Lessons Learned

- Individual detection rules are insufficient for multi-stage attacks without correlation
- Ransomware precursors can be identified before encryption if behavioral patterns are recognized
- L1 analyst intuition played a critical role in early escalation
- Detection engineering must focus on sequence-based behavior, not isolated commands

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Action Items

Action	**Owner**	**Priority**	**Target Date**
Deploy improved multi-event ransomware correlation rule in Sentinel	Nisha Kapoor (L3)	Critical	24-Apr-2026
Conduct environment-wide scan for similar PowerShell encoded executions	Rohan Desai (L2)	High	22-Apr-2026

Enforce stricter macro execution policies via Microsoft Defender for Office 365	Client IT Security Team	High	30-Apr-2026
Implement user awareness training focused on malicious document handling	Client HR + Security	Medium	15-May-2026
Review and restrict local admin privileges across engineering endpoints	Client IT Infrastructure Team	High	05-May-2026

Final Status: Contained — No Impact
Report Prepared By: SOC Team (L1–L3)
Reviewed By: MSSP Security Operations Lead